

$$1) \quad a - \frac{1}{2}x = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix}$$

$$A \cdot \frac{1}{2}x = \begin{pmatrix} -1 & 1 & 0 \\ 2 & -1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix} \\ = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

✓

$$b - b^T = (0 \ 0 \ 1)$$

$$A b = \begin{pmatrix} -1 & 1 & 0 \\ 2 & -1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \\ = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$b^T A b = (0 \ 0 \ 1) \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$= 1$$

✓

$$c - Ax = b$$

$$\Leftrightarrow x = \frac{b}{A}$$

$$\Leftrightarrow x = b \underbrace{\frac{1}{A}}$$

exist $\rightarrow$ vimos na linha a)

Na linha b) faremos que:

$$Ax = b$$

logo

$$\Leftrightarrow x = Ax \times \frac{1}{A}$$

$$\Leftrightarrow x = b \times 1 = b$$

(V)

$$d - A^T = \begin{pmatrix} -1 & 2 & 1 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

É a primeira linha de $(A^T)^{-1} = x$

$$A^T \cdot u = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\Leftrightarrow u = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \cdot (A^T)^{-1}$$

$$\Leftrightarrow u = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \cdot (A^{-1})^T$$

$$\Leftrightarrow u = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix}^T$$

$$\Leftrightarrow u = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 & -1 \\ 1 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$2) \quad a - A = \begin{pmatrix} 0 & 1 & 1 & 1 \\ -2 & 2 & 0 & 2 \\ -2 & 3 & 1 & 3 \end{pmatrix} \xrightarrow[L_2 \leftarrow L_2 - L_3]{OL_1} \begin{pmatrix} 0 & 1 & 1 & 1 \\ 0 & -1 & -1 & -1 \\ -2 & 3 & 1 & 3 \end{pmatrix} \xrightarrow[L_3 \leftarrow L_1]{OL_2} \begin{pmatrix} -2 & 3 & 1 & 3 \\ 0 & -1 & -1 & -1 \\ 0 & 1 & 1 & -1 \end{pmatrix}$$

$$\xrightarrow[L_3 \leftarrow L_3 + L_2]{OL_3} \begin{pmatrix} -2 & 3 & 1 & 3 \\ 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & -2 \end{pmatrix}$$

⑤

b - v

$$c - \begin{cases} x_2 + x_3 + x_4 = 0 \\ -2x_1 + 2x_2 + 2x_4 = 0 \\ -2x_1 + 3x_2 + x_3 + 3x_4 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} \underline{x_2} = -x_3 - x_4 \\ \underline{\quad} \end{cases} \Rightarrow \begin{cases} \underline{-2x_1 + 2(-x_3 - x_4) + 2x_4 = 0} \\ \underline{-2x_1 + 3(-x_3 - x_4) + x_3 + 3x_4 = 0} \end{cases}$$

$$\Rightarrow \begin{cases} \underline{-2x_1 - 2x_3 = 0} \\ \underline{-2x_1 - 2x_3 = 0} \end{cases} \Rightarrow \begin{cases} \underline{x_3} = -x_1 \\ \underline{\quad} \end{cases}$$

$$(2) \begin{cases} \underline{x_2 = -x_1 - x_4} \\ \underline{\quad} \end{cases}$$

Seja $x_3 = \alpha$ e $x_4 = \beta$

$(-\alpha, -\alpha - \beta, \alpha, \beta)$ v

d- reduzir a matriz A a escada reduzida.

$$A = \begin{pmatrix} 0 & 1 & 1 & 1 \\ -2 & 2 & 0 & 2 \\ -2 & 3 & 1 & 3 \end{pmatrix} \xrightarrow[\substack{L_2 \leftarrow L_2 - \frac{3}{2}L_3}]{OL_1} \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & -1/4 & -3/2 & -1/4 \\ -2 & 3 & 1 & 3 \end{pmatrix}$$

$$\xrightarrow[\substack{L_3 \leftarrow L_3 + 2L_2}]{OL_2} \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & -1/4 & -3/2 & -1/4 \\ 0 & 5/2 & -2 & 5/2 \end{pmatrix} \xrightarrow[\substack{L_1 \leftrightarrow L_2}]{OL_3} \begin{pmatrix} 1 & -1/4 & -3/2 & -1/4 \\ 0 & 1 & 1 & 1 \\ 0 & 5/2 & -2 & 5/2 \end{pmatrix}$$

$$\xrightarrow[\substack{L_1 \leftarrow L_1 + \frac{1}{4}L_2}]{OL_4} \begin{pmatrix} 1 & 0 & -5/4 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & -9/2 & 0 \end{pmatrix} \xrightarrow[\substack{L_3 \leftarrow L_3 + \frac{11}{2}L_2}]{OL_5} \begin{pmatrix} 1 & 0 & -5/4 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 11/2 \end{pmatrix}$$

$$L_3 \leftarrow L_3 - \frac{5}{2}L_2$$

F $\leftarrow$ E' $\leftarrow$ $\begin{matrix} \text{escada} \\ \text{reduzida} \end{matrix}$

$$\begin{aligned}
 3) \quad a - \det(2A^T A^{-1}) + \det(A^2) \\
 &= \det(2A^T) \det(A^{-1}) + \det(A^2) \\
 &= 2^5 \times 3 \times \frac{1}{3} + 3^2 \\
 &= 41 \quad \textcircled{F}
 \end{aligned}$$

b - V

c - V

d - F

e - S A fosse antisimétrica
 $A^T = -A$

$$\text{logo } \det(A)^T = -\det(A)$$

$$(i) \quad 3 = -3 \quad F$$

logo $\bar{m}$ não é antisimétrica

$\textcircled{V}$

$$\begin{aligned}
 4) \quad a - \begin{pmatrix} -1 & \alpha & 0 & 0 \\ 0 & -1 & 0 & \alpha \\ 1 & 0 & -\alpha & 0 \\ 0 & 0 & -\alpha & 1 \end{pmatrix} &\rightarrow \begin{pmatrix} -1 & \alpha & 0 & 0 \\ 0 & -1 & 0 & \alpha \\ 0 & \alpha & -\alpha & 0 \\ 0 & 0 & -\alpha & 1 \end{pmatrix} \\
 &\rightarrow \begin{pmatrix} -1 & \alpha & 0 & 0 \\ 0 & -\alpha & 0 & \alpha^2 \\ 0 & \alpha & -\alpha & 0 \\ 0 & 0 & -\alpha & 1 \end{pmatrix} \rightarrow \begin{pmatrix} -1 & \alpha & 0 & 0 \\ 0 & -\alpha & 0 & -\alpha^2 \\ 0 & 0 & -\alpha & \alpha^2 \\ 0 & 0 & -\alpha & 1 \end{pmatrix} \\
 &\rightarrow \begin{pmatrix} -1 & \alpha & 0 & 0 \\ 0 & -\alpha & 0 & -\alpha^2 \\ 0 & 0 & -\alpha & \alpha^2 \\ 0 & 0 & 0 & 1+\alpha^2 \end{pmatrix} \quad \textcircled{F}
 \end{aligned}$$

$$b- \begin{vmatrix} -1 & 2 & 0 & 0 \\ 0 & -1 & 0 & 2 \\ 1 & 0 & -2 & 0 \\ 0 & 0 & -2 & 1 \end{vmatrix} = (-1)^2 \times (-1) \times \begin{vmatrix} -1 & 0 & 2 \\ 0 & -2 & 0 \\ 0 & -2 & 1 \end{vmatrix} \\ + (-1)^3 \times 2 \times \begin{vmatrix} 0 & 0 & 2 \\ 1 & -2 & 0 \\ 0 & -2 & 1 \end{vmatrix}$$

$$= - \left\{ \begin{vmatrix} -1 & 0 & 2 \\ 0 & -2 & 0 \\ 0 & -2 & 1 \end{vmatrix} - 2 \begin{vmatrix} 0 & 0 & 2 \\ 1 & -2 & 0 \\ 0 & -2 & 1 \end{vmatrix} \right\} \begin{matrix} 0 \times (-2) \times 1 + 2 \times (-2) \times 1 + 0 \times 0 \times 0 - 2 \times (-2) \times 0 \\ - 0 \times (-2) \times 0 - 1 \times 1 \times 0 \\ = -4 \end{matrix}$$

$$\begin{matrix} (-1) \times (-2) \times 1 + 2 \times (-2) \times 0 + 0 \times 0 \times 0 \\ - 2 \times (-2) \times 0 - (-1) \times (-2) \times 0 - 0 \times 1 \times 0 \end{matrix}$$

$$= 2$$

$$= -2 + 8 = 6$$

(F)

$$c- (-1)^4 \times \begin{vmatrix} 0 & -1 & \alpha \\ 1 & 0 & \alpha \\ 0 & 0 & 1 \end{vmatrix} = 1$$

$$\begin{matrix} 0 \times 0 \times 1 + \alpha \times 0 \times 1 + \alpha \times 0 \times (-1) \\ - \alpha \times 0 \times 0 - 0 \times 0 \times \alpha - 1 \times 1 \times (-1) \end{matrix}$$

(F)

$$d- \left(\begin{array}{cccc|cccc} -1 & 2 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 2 & 0 & 1 & 0 & 0 \\ 1 & 0 & -2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & 1 & 0 & 0 & 0 & 1 \end{array} \right)$$

$$\xrightarrow[\substack{L_3 \leftarrow L_3 + L_1}]{\substack{OL_1 \\ \rightarrow}} \left(\begin{array}{cccc|cccc} -1 & 2 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 2 & -2 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & -2 & 1 & 0 & 0 & 0 & 1 \end{array} \right)$$

$$\begin{array}{l} \xrightarrow{OL_2} \\ L_1 \leftarrow L_1 - L_3 \end{array} \left(\begin{array}{cccc|cccc} -1 & 0 & 2 & 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 2 & -2 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -2 & 1 & 0 & 0 & 0 & 1 \end{array} \right)$$

$$\begin{array}{l} \xrightarrow{OL_4} \\ L_3 \leftarrow L_3 + L_2 \end{array} \left(\begin{array}{cccc|cccc} -1 & 0 & 2 & 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 1 & -2 & 2 & 1 & 1 & 1 & 0 \\ 0 & 0 & -2 & 1 & 0 & 0 & 0 & 1 \end{array} \right)$$

$$\begin{array}{l} \xrightarrow{OL_5} \\ L_3 \leftarrow L_3 + L_2 \end{array} \left(\begin{array}{cccc|cccc} -1 & 0 & 2 & 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 2 & 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 & 0 & 0 & 0 & 1 \end{array} \right)$$

$$\begin{array}{l} \xrightarrow{OL_6} \\ L_1 \leftarrow L_1 + L_3 \end{array} \left(\begin{array}{cccc|cccc} -1 & 0 & 0 & 2 & 1 & 2 & 0 & 0 \\ 0 & -1 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 2 & 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 & 0 & 0 & 0 & 1 \end{array} \right)$$

$$\begin{array}{l} \xrightarrow{OL_7} \\ L_3 \leftarrow -\frac{1}{2}L_3 \end{array} \left(\begin{array}{cccc|cccc} -1 & 0 & 0 & 2 & 1 & 2 & 0 & 0 \\ 0 & -1 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -\frac{1}{2} & -1 & -\frac{1}{2} & 0 \\ 0 & 0 & -2 & 1 & 0 & 0 & 0 & 1 \end{array} \right)$$

$$\begin{array}{l} \xrightarrow{OL_8} \\ L_4 \leftarrow L_4 + 2L_3 \end{array} \left(\begin{array}{cccc|cccc} -1 & 0 & 0 & 2 & 1 & 2 & 0 & 0 \\ 0 & -1 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -\frac{1}{2} & -1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & -1 & -1 & -2 & -1 & 1 \end{array} \right)$$

$$\begin{array}{l} \xrightarrow{OL_9} \\ L_3 \leftarrow L_3 - L_4 \\ L_2 \leftarrow L_2 + 2L_4 \\ L_1 \leftarrow L_1 + 2L_4 \end{array} \left(\begin{array}{cccc|cccc} -1 & 0 & 0 & 0 & -1 & -2 & -2 & 2 \\ 0 & -1 & 0 & 0 & -2 & -3 & -2 & 2 \\ 0 & 0 & 1 & 0 & \frac{1}{2} & 1 & \frac{1}{2} & -1 \\ 0 & 0 & 0 & -1 & -1 & -2 & -1 & 1 \end{array} \right)$$

$$\begin{array}{l} \xrightarrow{OL_1} \\ L_1 \leftarrow -L_1 \\ L_2 \leftarrow -L_2 \\ L_4 \leftarrow -L_4 \end{array} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 2 & 2 & -2 \\ 0 & 1 & 0 & 0 & 2 & 3 & 2 & -2 \\ 0 & 0 & 1 & 0 & 1/2 & 1 & 1/2 & -1 \\ 0 & 0 & 0 & 1 & 1 & 2 & 1 & -1 \end{array} \right)$$

* Erster map

e-f

II

$$1) a- \left(\begin{array}{ccc|c} 0 & 1 & 2\alpha & 0 \\ 2 & \beta-4 & 3\alpha & \alpha+1 \\ 2 & 0 & 3\alpha & 3 \end{array} \right)$$

$$\begin{array}{l} \xrightarrow{OL_1} \\ L_2 \leftrightarrow L_1 \end{array} \left(\begin{array}{ccc|c} 2 & \beta-4 & 3\alpha & \alpha+1 \\ 0 & 1 & 2\alpha & 0 \\ 2 & 0 & 3\alpha & 3 \end{array} \right) \xrightarrow{OL_2} \left(\begin{array}{ccc|c} 2 & \beta-4 & 3\alpha & \alpha+1 \\ 0 & 1 & 2\alpha & 0 \\ 0 & -\beta+4 & 0 & -\alpha+2 \end{array} \right)$$

$$\begin{array}{l} \xrightarrow{OL_3} \\ L_3 \leftarrow L_3 - (-\beta+4)L_2 \end{array} \left(\begin{array}{ccc|c} 2 & \beta-4 & 3\alpha & \alpha+1 \\ 0 & 1 & 2\alpha & 0 \\ 0 & 0 & -2\alpha(-\beta+4) & -\alpha+2 \end{array} \right)$$

$$b- \quad -2\alpha \neq 0$$

$$-\beta+4 \neq 0$$

$$\alpha \neq 0$$

$$\wedge \quad \beta \neq 4$$

P.D.

P.I.

$$\alpha = 0$$

$\vee$

$$\beta = 4$$

$\wedge$

$$\alpha = 2$$

P. Imd.

$$\alpha \neq 2 \quad \wedge \quad \beta = 4$$

$$c - \left(\begin{array}{ccc|c} 0 & 1 & 4 & 0 \\ 2 & 0 & 6 & 3 \\ 2 & 0 & 6 & 3 \end{array} \right)$$

.

$$d - \left| \begin{array}{ccc} 0 & 1 & 2 \\ 2 & -2 & 3 \\ 2 & 0 & 3 \end{array} \right| = 0 \times (-2) \times 3 + 2 \times 0 \times 2 + 3 \times 2 \times 1 \\ - 2 \times (-2) \times 2 - 0 \times 0 \times 3 - 2 \times 3 \times 1 \\ = \cancel{6} + 8 - \cancel{6} = 8$$

logo e' invariant